

Ex.No.7 - Analog Beamforming Analysis Using MATLAB

Objective – 1 Analyze Beamforming Gain (dB)

MATLAB Code:

```
clc;

clear;

close all;

antennas = [2 4 8 16 32];

gain = 10*log10(antennas);

figure

bar(antennas, gain)

grid on

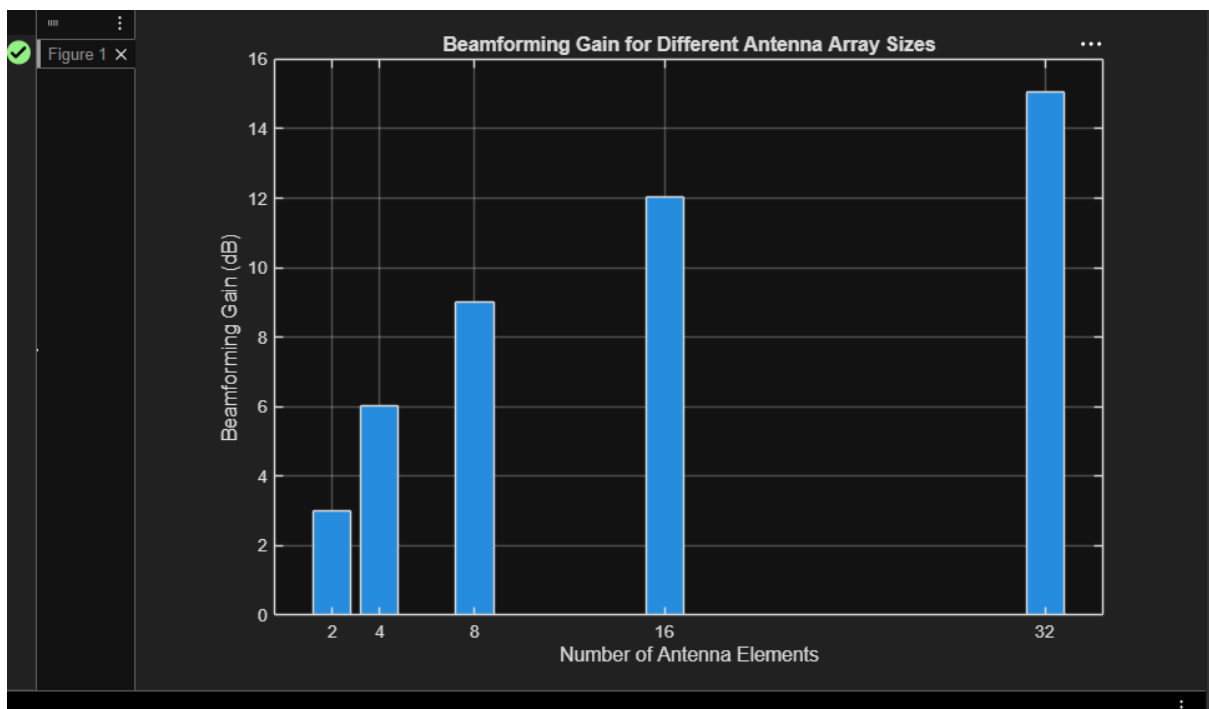
% Number of Antenna Elements

% Beamforming Gain (dB)

xlabel('Number of Antenna Elements')

ylabel('Beamforming Gain (dB)')

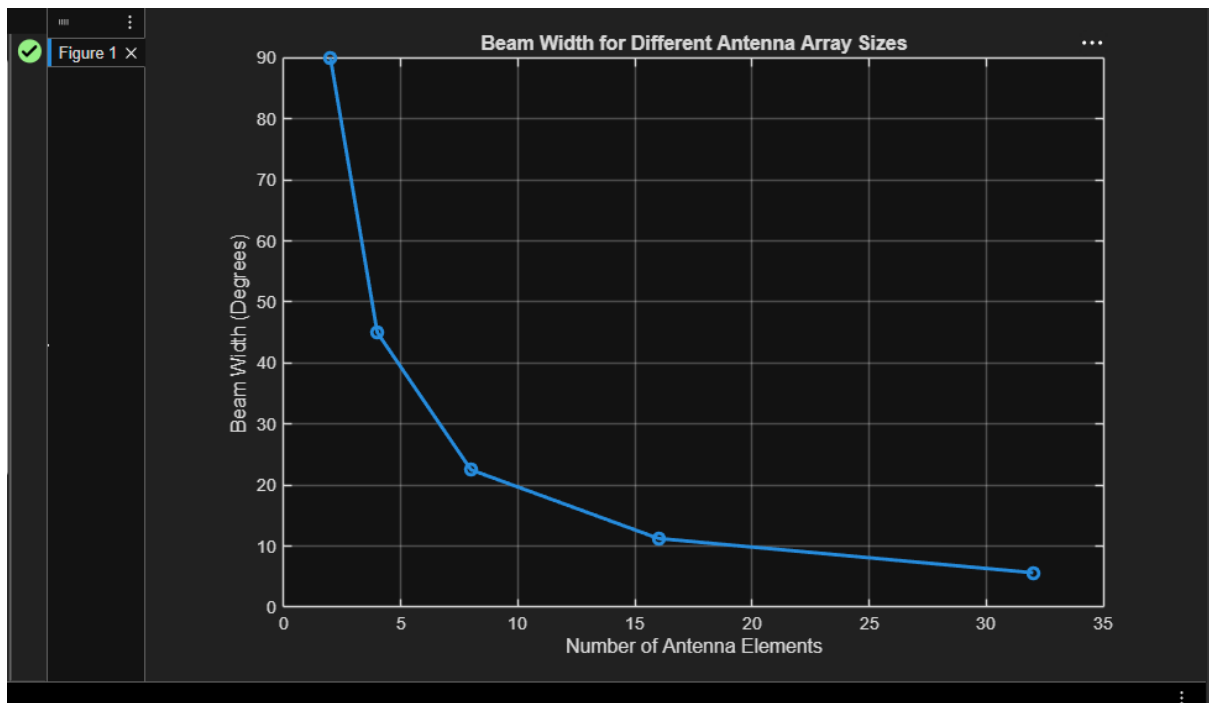
title('Beamforming Gain for Different Antenna Array Sizes')
```



Objective – 2 Compare Beam Width (Degrees)

MATLAB Code:

```
clc;  
  
clear;  
  
close all;  
  
antennas = [2 4 8 16 32];  
  
% Number of Antenna Elements  
  
beamwidth = [90 45 22.5 11.25 5.625];  
  
figure  
  
% Beam Width (degrees)  
  
plot(antennas, beamwidth,'o-','LineWidth',2)  
  
grid on  
  
xlabel('Number of Antenna Elements')  
  
ylabel('Beam Width (Degrees)')  
  
title('Beam Width for Different Antenna Array Sizes')
```



Objective – 3 Analyze Main Lobe Direction (Degree)

MATLAB Code:

```
clc;
```

```
clear;
```

```
close all;
```

```
theta = [-60 -30 0 30 60];
```

```
gain = [1 1 1 1 1];
```

```
figure
```

```
% Main Lobe Direction (degrees)
```

```
% Constant normalized gain
```

```
polarplot(deg2rad(theta), gain, 'o', 'LineWidth', 2)
```

```
title('Main Lobe Direction in Analog Beamforming')
```

